SHUBHAJIT NANDI

Senior Backend Engineer | Applied AI & Voice Platforms

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SUMMARY

Senior backend engineer with 11+ years across telecom, cybersecurity, and voice platforms, including LLM and voice integrations. Build AI content-generation workflows with structured outputs, tool calling, and operational controls. Lead five engineers delivering a Spring Boot SaaS platform with responsibility for backend architecture and cloud delivery.

TECHNICAL SKILLS

AI Integration: OpenAI and Gemini APIs, structured outputs, tool calling, timeouts, retries, rate-limit handling, token-cost monitoring, caching, tracing

Voice & Communications: ElevenLabs, Twilio, LiveKit, SIP, Asterisk

Backend & Messaging: Java, Python, Spring Boot, FastAPI, Kafka, Kafka Connect, NATS JetStream, CQRS

Cloud & Delivery: AWS, Terraform, Docker, Kubernetes, GitHub Actions, Amazon S3, CloudFront

Data & Performance: PostgreSQL, MongoDB, Redis, Grafana, Grafana k6, load testing

PROFESSIONAL EXPERIENCE

Aspire Tech Services and Solutions Corp. (US-based client)

Senior Software Engineer | Jun 2025 - Present | Remote, Dhaka

- Lead a five-engineer team delivering ASAT-V2, a security-awareness SaaS platform, owning backend architecture, Spring Boot services, and production delivery.
- Built LLM content-generation workflows with structured outputs and tool calling, handling timeouts, retries, and rate limits for customer-specific phishing simulations.
- Manage latency, token cost, caching, and tracing for AI integrations in ASAT-V2.
- Built email, SMS, and voice phishing simulations, integrating Twilio and ElevenLabs and capturing participant responses for security-awareness risk reporting.
- Reduced JavaScript asset loading time by approximately 50% by moving asset delivery to Amazon S3 and CloudFront.
- Own deployment environments from server setup through releases, implementing GitHub Actions pipelines for container builds, automated deployments, and versioned rollback.
- Mentored frontend engineers on micro-frontend design and implementation, aligning backend APIs and UI workflows across phishing, billing, and training modules.

Verbex

Senior Software Engineer | Aug 2023 - Jun 2025 | Dhaka, Bangladesh

- Built seven Spring Boot services for a conversational voice AI platform using Java 21, Axon Framework, and CQRS, covering agent management, call orchestration, and transcription analysis.
- Integrated LiveKit SIP trunking to support inbound and outbound calls for the conversational AI platform.
- Built a Kafka and Kafka Connect data-warehousing pipeline, reducing processing latency by approximately 50% for downstream analytics.
- Developed a custom Kafka Connect transformation plugin to reshape complex source data for BI dashboards.

Dotlines Bangladesh

Senior Software Engineer | Nov 2020 - Aug 2023 | Dhaka, Bangladesh

- Developed an asynchronous outbound-call gateway using Python, FastAPI, asyncio, and NATS JetStream to increase parallel call-processing capacity.
- Built a C-based SMSC using SMPP, SS7, and SCTP, handling 3,000 TPS and reducing per-message routing costs by 90% compared with third-party providers.
- Developed Java payment-gateway services deployed on Grameenphone and Banglalink platforms, integrating multiple third-party payment providers.

SSD-TECH

Software Engineer | Feb 2015 - Nov 2020 | Dhaka, Bangladesh

- Developed SMS delivery systems in C and PHP with MySQL and Redis, running on Banglalink's production platform at 2,000 TPS.
- Built reusable SMPP and SIP libraries in C for multiple telecom projects; implemented call-patching and conference-call features using Asterisk.

EDUCATION

Bachelor's in Computer Science and Engineering | 2010 - 2014

Chittagong University of Engineering and Technology (CUET), Chittagong, Bangladesh

CERTIFICATION & LANGUAGES

Splunk Certified Cybersecurity Defense Analyst

Bengali: Native | English: Fluent